STAT6611

- Lecture 13 (T, Oct 6)
- solutions posted to Homework 2
- Homework 3 assigned due F, Oct 16 by midnight
- Take home midterm to be given on R, Oct 15 & due F, Oct 23 (one week)
- We'll cancel class on R, Oct 15
- I'll fix our canvas front-page

Proof. Note that $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $X'X\underline{r} = \underline{\lambda}$ for some $\underline{r}$ (by Theorem 2.7). Thus, by Theorem 2.21, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if , for every $\underline{r}$, there exists a $\underline{q}$ such that

$$HX\underline{r} = X\underline{q}.$$

Need to show: for any $\underline{r} \in \mathbb{R}^p$, $HX\underline{r} = X\underline{q}$ holds for some $\underline{q}$ (depending on $\underline{r}$)

$\Leftrightarrow HX = XQ$ for some $p \times p$ matrix Q

$\Rightarrow$ let $\underline{r}_i$ be the ith column of $p \times p$ identity matrix I_p

& $HX\underline{r}_i = X\underline{q}_i$ for some $\underline{q}_i$, $i = 1, \ldots, p$.

Then, $HX = HXI_p = HX[\underline{r}_1, \ldots \underline{r}_p] = X[\underline{q}_1, \ldots, \underline{q}_p]$

$\Leftarrow$ Suppose $HX = XQ \Rightarrow HX\underline{r} = XQ\underline{r} = X\underline{q} = XQ$, $Q = [\underline{q}_1 \ldots \underline{q}_p]$

The next corollary gives a condition $HP_X = P_XH$ that is equivalent to $HX = XQ$ for some Q given in Corollary 1, where $P_X = X(X'X)^-X'$. Note that $(HP_X)' = P_X'H' = P_XH$ is not necessarily a symmetric matrix.

holds for any $\underline{r} \in \mathbb{R}^p$

$P_X \equiv X(X'X)^-X'$ projection matrix of X Note: $P_X' = P_X$, $H' = H$

Corollary 2. Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $HP_X = P_XH$ (that is, HP_X is a symmetric matrix).

$HP_X = P_XH$ holds $\Leftrightarrow HX = XQ$

$= (HP_X)'$

Proof. By Corollary 1, it suffices to show that $HP_X = P_XH$ (that is, HP_X is a symmetric matrix) if and only if $HX = XQ$ for some Q.

$\Rightarrow$ Suppose $HP_X = P_XH$, then multiply both sides by X

get $HP_XX = P_XHX$ or $HX = XQ$ for $Q = (X'X)^-X'HX$

HP_X X P_X $X(X'X)^-X'$ X So SLSE must be BLUE. (corollary 1)

$\Leftarrow$ If $HX = XQ$, then

$P_XHP_X = P_XHX(X'X)^-X' = P_XXQ(X'X)^-X' = XQ(X'X)^-X'$

symmetric since $P_X' = P_X$, $H' = X$ $= HX(X'X)^-X' = HP_X$

So HP_X is symmetric ie $HP_X = (HP_X)' = P_X H$

Remarks. Theorem 2.21 and its two corollaries are still valid if $\text{Var}(\underline{y}) = \sigma^2 H$ is put in place of H.

$\text{Var}(\underline{y}) = \text{Var}(\underline{\epsilon}) = V$ beyond $\sigma^2 H$

Furthermore, under Model VII with variance V:

1. Theorem 2.21 and its two corollaries are still valid if V is put in place of H.

2. That is, under Model VII, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ if and only if there exists a vector $\underline{q}$ such that $VX\underline{r} = X\underline{q}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$;

3. And, under Model VII, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists a Q such that $VX = XQ$ or if and only if $VP_X = P_X V$.

SLSE is BLUE iff $VX = XQ$ iff $(VP_X)' = P_X'V' = P_X V$

4. Above, elements of $\underline{q}$ and Q may be functions of $\underline{\theta}$.

5. For more details, see George Zyskind's paper, "On canonical forms, non-negative covariance matrices and best and simple least squares linear estimators in linear models" (*The Annals of Mathematical Statistics*, Volume 38, 1967, 1092–1109).

Example. Suppose that we have an observation on each of s variates for each of r "individuals." Let $\underline{y}_i = \begin{bmatrix} y_{i1} \\ \vdots \\ y_{ir} \end{bmatrix}$ represent the vector whose elements are the observations on the ith variate. Take the model equation for $\underline{y}_i$ to be

$$\underline{y}_i = U\underline{\beta}_i + \underline{\epsilon}_i, \qquad i = 1, \ldots, s$$

where, for some t, U is a given $r \times t$ matrix and $\underline{\beta}_i$ is a $t \times 1$ vector of unknown parameters.

Suppose that

$$\text{Var} \begin{bmatrix} y_{1j} \\ \vdots \\ y_{sj} \end{bmatrix} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1s} \\ \vdots & \vdots & \vdots \\ \sigma_{s1} & \cdots & \sigma_{ss} \end{bmatrix},$$

where some or all of the $\sigma_{ii'}$'s may be unknown parameters; and suppose that $\text{cov}(y_{ij}, y_{i'j'}) = 0$ for $i = 1, \ldots, s$, $i' = 1, \ldots, s$, $j = 1, \ldots, r$, $j' = 1, \ldots, r$, and $j \neq j'$ (that is, observations from the same individuals may be correlated, but observations from different individuals are uncorrelated).

This model can be viewed as a special case of Model VII, where

$$\underline{y} = \begin{bmatrix} \underline{y}_1 \\ \vdots \\ \underline{y}_s \end{bmatrix}, \; \underline{\epsilon} = \begin{bmatrix} \underline{\epsilon}_1 \\ \vdots \\ \underline{\epsilon}_s \end{bmatrix}, \; \underline{\beta} = \begin{bmatrix} \underline{\beta}_1 \\ \vdots \\ \underline{\beta}_s \end{bmatrix}, \; X = \begin{bmatrix} U & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & U \end{bmatrix}, \text{ and } V = \begin{bmatrix} \sigma_{11} I_r & \cdots & \sigma_{1s} I_r \\ \vdots & \vdots & \vdots \\ \sigma_{s1} I_r & \cdots & \sigma_{ss} I_r \end{bmatrix},$$

and where the elements of $\underline{\theta}$ consist of the unknown $\sigma_{ii'}$'s.

Here, the (i, i')th block of V is

$$V_{ii'} = \mathrm{cov}(\underline{\epsilon}_i, \underline{\epsilon}_{i'}) = \mathrm{cov}\left(\begin{bmatrix} \epsilon_{i1} \\ \vdots \\ \epsilon_{ir} \end{bmatrix}, \begin{bmatrix} \epsilon_{i'1} \\ \vdots \\ \epsilon_{i'r} \end{bmatrix} \right) = \begin{bmatrix} \mathrm{cov}(\epsilon_{i1}, \epsilon_{i'1}) & \cdots & \mathrm{cov}(\epsilon_{i1}, \epsilon_{i'r}) \\ \vdots & \vdots & \vdots \\ \mathrm{cov}(\epsilon_{ir}, \epsilon_{i'1}) & \cdots & \mathrm{cov}(\epsilon_{ir}, \epsilon_{i'r}) \end{bmatrix}$$

$$= \begin{bmatrix} \sigma_{ii'} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_{ii'} \end{bmatrix} = \sigma_{ii'} I_r.$$

Suppose that we wish to estimate an estimable function $\underline{\lambda}'\underline{\beta} = \sum_{i=1}^{s} \underline{\lambda}_i'\underline{\beta}_i$ [where $\underline{\lambda}' = (\underline{\lambda}_1', \ldots, \underline{\lambda}_s')$] of the elements of the parameter vector $\underline{\beta}$. Consider the estimator obtained by carrying out a separate least squares analysis on each variate, that is, the estimator $\sum_{i=1}^{s} \underline{\lambda}_i'\widetilde{\underline{\beta}}_i$, where $U'U\widetilde{\underline{\beta}}_i = U'\underline{y}_i$.

or $\sum_{i=1}^{s} \underline{\lambda}_i'(U'U)^{-}U'\underline{y}_i$

$(U'U)^{-}U'\underline{y}_i$ is solution to $U'U\widetilde{\underline{\beta}}_i = U'\underline{y}_i$

Question. Is the estimator $\sum_{i=1}^{s} \underline{\lambda}_i'\widetilde{\underline{\beta}}_i$ a BLUE of $\underline{\lambda}'\underline{\beta}$ based on all the observations on all individuals?

Note that the normal equations $X'X\widetilde{\underline{\beta}} = X'\underline{y}$ based on all the observations on all individuals can be written as

$$\begin{bmatrix} U'U & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & U'U \end{bmatrix} \begin{bmatrix} \widetilde{\underline{\beta}}_1 \\ \vdots \\ \widetilde{\underline{\beta}}_s \end{bmatrix} = \begin{bmatrix} U'\underline{y}_1 \\ \vdots \\ U'\underline{y}_s \end{bmatrix}, \quad \text{or} \quad U'U\widetilde{\underline{\beta}}_i = U'\underline{y}_i,\ i = 1, \ldots, s.$$

so, $\underline{\lambda}'\widetilde{\underline{\beta}} = \sum_{i=1}^{s} \underline{\lambda}_i'\widetilde{\underline{\beta}}_i$ $\quad \widetilde{\underline{\beta}}_i = (U'U)^{-}U'\underline{y}_i$

Thus, the above question is equivalent to the question of whether the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$. We have

$$VX = \begin{bmatrix} \sigma_{11}I_r & \cdots & \sigma_{1s}I_r \\ \vdots & \vdots & \vdots \\ \sigma_{s1}I_r & \cdots & \sigma_{ss}I_r \end{bmatrix} \begin{bmatrix} U & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & U \end{bmatrix} = \begin{bmatrix} \sigma_{11}U & \sigma_{12}U & \cdots & \sigma_{1s}U \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{s1}U & \sigma_{s2}U & \cdots & \sigma_{ss}U \end{bmatrix}$$

$$= \begin{bmatrix} U & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & U \end{bmatrix} \begin{bmatrix} \sigma_{11}I_t & \sigma_{12}I_t & \cdots & \sigma_{1s}I_t \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{s1}I_t & \sigma_{s2}I_t & \cdots & \sigma_{ss}I_t \end{bmatrix} = XQ$$

for

$$Q = \begin{bmatrix} \sigma_{11}I_t & \sigma_{12}I_t & \cdots & \sigma_{1s}I_t \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{s1}I_t & \sigma_{s2}I_t & \cdots & \sigma_{ss}I_t \end{bmatrix},$$

so that the SLSE of any estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$.

2.12 Best Linear Unbiased Prediction

Suppose that the model for the data vector $\underline{y} = X\underline{\beta} + \underline{\epsilon}$ is Model III, with H nonsingular, and that we wish to predict (or estimate)

$$\underline{\lambda}'\underline{\beta} + u,$$

where $u \in \mathbb{R}$ is a random variable having zero expectation $E(u) = 0$ and finite variance. For example, $\underline{\lambda}'\underline{\beta} + u$ might correspond to a future data point.

A linear predictor (or estimator) $c + \underline{a}'\underline{y}$ of $\underline{\lambda}'\underline{\beta} + u$ is said to be unbiased if

$$E(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta} \quad \text{for all} \quad \underline{\beta},$$

that is, if $E(c + \underline{a}'\underline{y}) = E(\underline{\lambda}'\underline{\beta} + u)$ for all $\underline{\beta}$. If there exists a linear unbiased predictor (or estimator) of $\underline{\lambda}'\underline{\beta} + u$, then $\underline{\lambda}'\underline{\beta} + u$ is said to be predictable (or estimable).

Suppose $\text{cov}(\underline{\epsilon}, u) = \sigma^2 \underline{k}$ for some given vector $\underline{k}$, and $\underline{\lambda}'\underline{\beta}$ is estimable.

One way to motivate predictors is to pretend (though this need not be true) that $(\underline{\epsilon}, u)$ are jointly normal: if so, the conditional expectation of u given $\underline{\epsilon}$ would be

$$E(u|\underline{\epsilon}) = \underline{k}'H^{-1}\underline{\epsilon}.$$

In predicting $\underline{\lambda}'\underline{\beta} + u$, we do not know $\underline{\beta}$ or $\underline{\epsilon}$, but we could estimate these.

Let $\widehat{\underline{\beta}} = (X'H^{-1}X)^{-}X'H^{-1}\underline{y}$ be a solution to $X'H^{-1}X\widehat{\underline{\beta}} = X'H^{-1}\underline{y}$.

Then, estimate $\underline{\lambda}'\underline{\beta}$ by $\underline{\lambda}'\widehat{\underline{\beta}}$ and predict u by $\widehat{u} \equiv \underline{k}'H^{-1}\widehat{\underline{\epsilon}}$ using residuals $\widehat{\underline{\epsilon}} \equiv \underline{y} - X\widehat{\underline{\beta}}$.

That is, consider predicting $\underline{\lambda}'\underline{\beta} + u$ by

$$\begin{aligned}
\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u} &\equiv \underline{\lambda}'\widehat{\underline{\beta}} + \underline{k}'H^{-1}\widehat{\underline{\epsilon}} \\
&\equiv \underline{\lambda}'\widehat{\underline{\beta}} + \underline{k}'H^{-1}(\underline{y} - X\widehat{\underline{\beta}}) \\
&= \underline{k}'H^{-1}\underline{y} + (\underline{\lambda}' - \underline{k}'H^{-1}X)\widehat{\underline{\beta}} \\
&= \left[\underline{k}'H^{-1} + (\underline{\lambda}' - \underline{k}'H^{-1}X)(X'H^{-1}X)^{-}X'H^{-1}\right]\underline{y} \equiv \underline{b}'\underline{y},
\end{aligned}$$

where

$$\underline{b} \equiv H^{-1}\underline{k} + H^{-1}X\left[(X'H^{-1}X)^{-}\right]'(\underline{\lambda} - X'H^{-1}\underline{k}).$$

By Theorem 2.19, we have

$$E(\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u}) = E(\underline{\lambda}'\widehat{\underline{\beta}}) + \underline{k}'H^{-1}\left[E(\underline{y}) - E(X\widehat{\underline{\beta}})\right] = \underline{\lambda}'\underline{\beta} + \underline{k}'H^{-1}(X\underline{\beta} - X\underline{\beta}) = \underline{\lambda}'\underline{\beta}.$$

Thus, $\underline{b}'\underline{y} = \underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u}$ is a linear unbiased predictor of $\underline{\lambda}'\underline{\beta} + u$.

Let $c + \underline{a}'\underline{y}$ be any linear unbiased predictor of $\underline{\lambda}'\underline{\beta} + u$, requiring $\underline{a}'X = \underline{\lambda}'$ and $c = 0$.

One can show that mean squared error of $c + \underline{a}'\underline{y}$ in prediction cannot be smaller than that of $\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u}$, that is

$$E\left[c + \underline{a}'\underline{y} - (\underline{\lambda}'\underline{\beta} + u)\right]^2 \geq E\left[\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u} - (\underline{\lambda}'\underline{\beta} + u)\right]^2$$

Because $c + \underline{a}'\underline{y}$ and $\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u}$ are both linear unbiased predictors,

$$E\left[c + \underline{a}'\underline{y} - (\underline{\lambda}'\underline{\beta} + u)\right]^2 = \text{Var}\left[c + \underline{a}'\underline{y} - (\underline{\lambda}'\underline{\beta} + u)\right] = \text{Var}\left(\underline{a}'\underline{y} - u\right)$$

and

$$E\left[\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u} - (\underline{\lambda}'\underline{\beta} + u)\right]^2 = \text{Var}\left[\underline{b}'\underline{y} - (\underline{\lambda}'\underline{\beta} + u)\right] = \text{Var}\left(\underline{b}'\underline{y} - u\right)$$

Write

$$\begin{aligned}\text{Var}\left(\underline{a}'\underline{y} - u\right) &= \text{Var}\left[(\underline{b}'\underline{y} - u) + (\underline{a} - \underline{b})'\underline{y}\right] \\ &= \text{Var}\left(\underline{b}'\underline{y} - u\right) + \text{Var}\left[(\underline{a} - \underline{b})'\underline{y}\right] + 2\text{cov}\left[(\underline{a} - \underline{b})'\underline{y}, \underline{b}'\underline{y} - u\right].\end{aligned}$$

Since $\text{cov}\left((\underline{a}-\underline{b})'\underline{y}, \underline{b}'\underline{y} - u\right)$

$= \text{cov}\left((\underline{a}-\underline{b})'\underline{y}, \underline{b}'\underline{y}\right) - \text{cov}\left((\underline{a}-\underline{b})'\underline{y}, u\right)$

$= (\underline{a}-\underline{b})'H\sigma^2\underline{b} - (\underline{a}-\underline{b})'\underline{k}\sigma^2$

$= \sigma^2(\underline{a}-\underline{b})'[H\underline{b} - \underline{k}]$

$= \sigma^2(\underline{a}-\underline{b})'[X J]$

$= \sigma^2(X'\underline{a} - X'\underline{b})'J = 0$

($X'\underline{a} = \underline{\lambda}$, $X'\underline{b} = \underline{\lambda}$)

Therefore, among linear unbiased predictors of $\underline{\lambda}'\underline{\beta} + u$, the predictor $\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u}$ has uniformly smaller mean squared error than any other linear unbiased predictor. That is, $\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u}$ is the unique best linear unbiased predictor (BLUP) of $\underline{\lambda}'\underline{\beta} + u$.